Machine Learning – Fundamental Concepts

BMED365: Topic List M01–M10

BMED365

Spring 2026

Overview

1. **What is machine learning?**
   - M01: Definition and distinction from traditional programming
   - M02: Supervised vs. Unsupervised learning
   - M03: Features and Labels
2. **Validation and Model Types**
   - M04–M05: Training/test set, overfitting/underfitting
   - M06–M07: Bias-Variance trade-off, K-fold cross-validation
   - M08–M10: Baseline, classification vs. regression, simple models

M01: Define machine learning and distinguish from traditional programming

Traditional programming:

- Explicit rules defined by programmer
- `if-else` logic, deterministic
- Input + Rules → Output
- Requires full domain knowledge beforehand

Machine Learning:

- Learns rules from data automatically
- Pattern recognition, statistical
- Input + Output → Rules (model)
- Scales with data volume

Definition (Tom Mitchell, 1997)

"A computer program is said to ***learn from experience*** *E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E."*

Key insight

ML shifts focus from **programming explicit rules** to **learning patterns from data**. The algorithm improves its performance through experience.

M02: Supervised vs. Unsupervised learning

Supervised learning:

- Data has **labels** (ground truth)
- Model learns input→output mapping
- Types: Classification, regression
- Medical: Diagnosis, prognosis
- Requires manual annotation of data

Unsupervised learning:

- Data **without** labels
- Finds hidden patterns/structures
- Types: Clustering, dimensionality reduction
- Medical: Patient subgroups
- No ground truth – exploratory analysis

Semi-supervised – principles and applications

Combines few labeled examples with many unlabeled. Useful when annotation is expensive/time-consuming.

Principles: *Smoothness* – nearby data points often have the same label. *Tabular:* patients with similar lab values often have the same diagnosis. *Image:* MR slices with similar pixel intensities show similar pathology. *Cluster* – data groups naturally; points in the same cluster often belong to the same class. *Tabular:* diabetic patients cluster in feature space. *Image:* brain MRI of healthy vs. Alzheimer form separate clusters. *Manifold* – a 256×256 image = 65536 pixels, but meaningful variation is determined by few factors (anatomy, pathology). Images don't "fill" the entire space, but lie on a low-dim. manifold.

Applications: Medical image segmentation, text classification, speech recognition, drug discovery, genomics.

M03: Features (input) and Labels (output)

Definitions

- **Features** (X): Input variables / properties / characteristics that describe the data
- **Labels** (y): Target variable / what we want to predict (ground truth)

Example – Heart disease prediction (tabular data):

Age	Cholesterol	Blood Pressure	BMI	Disease?
55	240	140	28	Yes
32	180	120	23	No
67	210	155	31	Yes

Features X (columns = variables) **Label** y

Rows = observations/patients, Columns = variables → **Matrix**

M04: Why training set and test set?

The problem: How do we know if the model generalizes to new data?

The solution: Data splitting

1. Split the dataset into two (or three) parts
2. **Training set** (~70–80%): Train the model
3. **Test set** (~20–30%): Evaluate on unseen data
4. (Validation set: Tuning of hyperparameters)

Important

The model should **never** see the test data during training!

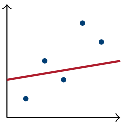
scikit-learn

```
train_test_split(X, y, test_size=0.2)
```

M05: Overfitting and Underfitting

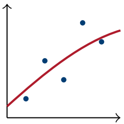
Underfitting:

- Too simple model
- High bias
- Poor on training and test



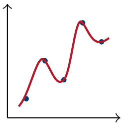
Good fit:

- Right complexity
- Balance
- Generalizes well



Overfitting:

- Too complex model
- High variance
- Learns noise in data



Signs of overfitting

Large difference between training and test performance. **Performance** = measure of how well the model performs (e.g. accuracy, F1).

Total prediction error:

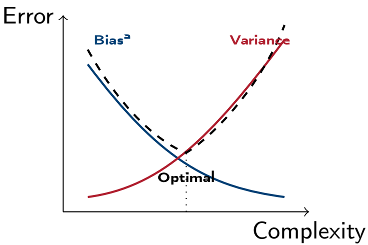
$$\text{MSE} = \underbrace{\text{Bias}^2}_{\text{systematic error}} + \underbrace{\text{Variance}}_{\text{random error}} + \underbrace{\sigma^2}_{\text{irreducible}}$$

Bias (systematic error):

- Error due to simplified model
- High bias → **underfitting**
- Model "misses" the pattern

Variance (random error):

- Sensitive to small changes in data
- High variance → **overfitting**
- Model "memorizes" noise



The goal

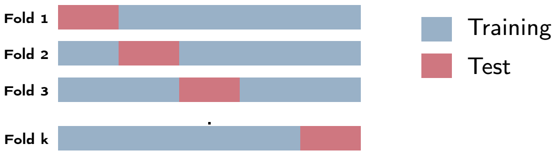
Find the balance that minimizes **total error** on new data. σ^2 = noise in data (cannot be reduced).

M07: K-fold cross-validation

Problem: A single train/test split can give random results

Solution: K-fold Cross-Validation

1. Split data into k equal parts ("folds")
2. For each fold: use it as test, rest as training
3. Average of all k evaluations



Advantages: More robust estimate, uses all data for both training and testing

M08: Baseline model

What is a baseline?

A **simple reference model** that our ML model must beat to be useful.

Examples of baselines:

- **Classification:** Always predict the majority class
  - Dataset with 90% healthy, 10% sick $\rightarrow$ baseline accuracy = 90%
- **Regression:** Always predict the mean value
- **Time series:** Use previous value as prediction

Why important?

90% accuracy sounds impressive, but not if baseline is 90%! Shows whether ML adds **actual value**. Reveals imbalanced datasets.

M09: Classification vs. Regression

Classification:

- Predicts **categories/classes**
- Discrete outcomes
- Examples:
  - Sick / Healthy
  - Cancer type A / B / C

Evaluation metrics:

- Accuracy, Precision, Recall
- F1-score, AUC-ROC

Regression:

- Predicts **continuous values**
- Numerical outcomes
- Examples:
  - Blood pressure
  - Drug dosage

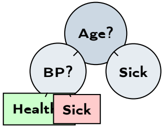
Evaluation metrics:

- MSE, RMSE, MAE
- R^2 (explained variance)

M10: Simple ML models

Decision tree:

- Series of yes/no questions
- Easy to interpret
- Can overfit

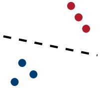


k-Nearest Neighbors:

- Find k most similar examples
- Majority voting
- Simple, but slow

Logistic regression:

- Linear boundary
- Gives probabilities
- Fast and interpretable



Tip

Always start with simple models before trying complex ones!

Summary: M01–M10

Key concepts:

- ML learns from data
- Supervised / Unsupervised / Semi-supervised
- Features (X), Labels (y)
- Training / test / validation
- Over-/underfitting

Best practices:

- Cross-validation
- Compare with baseline
- Classification $\neq$ Regression
- Start simple
- Bias-variance balance

Future: ML + GenAI

- LLM agents for data analysis
- Multimodal models (text+image)
- RAG + clinical guidelines
- Automated ML pipeline
- Human-in-the-loop AI

Medical relevance – ML applications in health, medicine and biotechnology

Diagnostics: Image diagnostics (radiology, pathology, ophthalmology, dermatology), biomarker detection, ECG/EEG analysis.
Prognosis: Overall survival (OS), progression-free survival (PFS), readmission, mortality risk, disease progression.
Therapy: Treatment selection, dose optimization, treatment response, radiation planning, surgical navigation.
Drug development: Drugability, target identification, ADMET prediction, virtual screening, de novo molecule design.
Healthcare system: Resource allocation, capacity planning, triage, waitlist optimization, outbreak detection.
Genomics/biotech: Variant classification, gene expression, protein structure (AlphaFold), single-cell, CRISPR off-target.
Psychiatry/neurology: Sentiment analysis, speech markers, neurodegeneration, seizure prediction.
Public health: Epidemiological modeling, vaccine strategy, social determinants, health economics (QALY/DALY).

Human-in-the-loop: **Patient image** → $\boxed{\text{ML: “possible malignant” (87\%)}}$ → **Radiologist evaluates** → **Confirms/rejects** → **Feedback** → **Model improves**